

## Assessment 2: Ontology, Prolog, Modal Logic

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**Ontology overview:** My ontology includes ten main classes: Sandwich, Bread, Filling, Spread, Vegetable, Cheese, Meat, Sauce, SandwichType, and PreparationMethod. These represent the different parts of a sandwich.

Sandwich is the “parent” concept in the ontology. All the other classes describe things that are part of a sandwich or help explain what kind of sandwich it is. Other ingredients like Spread, Vegetable, and Sauce are optional. I also added SandwichType and PreparationMethod to classify sandwiches by style, like grilled or club sandwiches.

To describe how these parts fit together, I created five properties: hasIngredient, hasFilling, hasBread, hasPreparationMethod, and hasSandwichType. These connect the class Sandwich to its ingredients or descriptions. I also added a data property called hasNumberOfFillings, which lets me define rules like “a sandwich must have two or more fillings.”

**Class hierarchy:** I created a small hierarchy to group related classes together. Filling has three subclasses: Meat, Cheese, and Vegetable. These are common sandwich fillings, and separating them helps a computer understand what kind of filling is being used. For example, if something is a Meat, it will also be understood as a Filling automatically because it inherits from that class.

I also thought about grouping Spread and Sauce under something like Additions, but to keep the ontology simple and easy to test, I didn’t include that abstract group formally. As for the properties, I could have made hasIngredient the “main” property and made hasFilling and hasBread into special cases of it. But I kept them separate so the ontology is easier to read and debug, especially for people who might be new to using tools like Protégé.

**Axiom explanation:** I included one axiom using number restrictions. It says:

$\text{Sandwich} \sqsubseteq (\geq 2 \text{ hasFilling.Filling})$

This means: a sandwich must have at least two fillings.

I chose this rule because the Wikipedia article mentions, “Some sandwiches have many layers of filling” and “A sandwich is two pieces of bread with something between them” (Simple English Wikipedia, 2023). By using the number “2” here, the ontology checks if a sandwich is properly made according to this structure.

Even though there might be arguments that a butter sandwich or a sandwich with one slice of cheese still counts, I used this rule to test the applying of logical restrictions in an ontology. If someone wants to create a simpler sandwich, the axiom can be changed or made more flexible in future versions.

**Advantages and Disadvantages of ontology:** One of the best things about building this ontology is that it helps turn everyday knowledge into something a computer can use. If I

connect this to an AI system, it could be used to suggest recipes, check dietary preferences, or even help a robot make a sandwich.

Another benefit is that this ontology could be combined with larger food-related ontologies like FoodOn or the BBC Food ontology. Since sandwiches can be made in so many ways, this structure allows lots of flexibility while still making logical sense.

But there are also a few challenges. Natural language is full of grey areas and cultural differences. For example, people have different opinions about whether a hot dog is a sandwich or not.

Also, as I try to include more ingredients, preparation styles, or global variations, the ontology might become too big and hard to manage. It could end up needing many subclasses, properties, and exceptions, which makes it less user-friendly.

Finally, setting a rule like “two fillings minimum” could block some sandwiches that are perfectly valid in the real world. That’s why it’s important to leave room for optional constraints or allow users to pick different rules depending on what the system is being used for.

#### **Acknowledgement:**

- **AI System:** GitHub Copilot
- **Publisher:** GitHub
- **URL:** <https://github.com/features/copilot>
- **Usage:** To help in writing the Prolog code and modal logic expressions for the ontology-based tasks.

**Summary:** This sandwich ontology is a small but useful example of how I can organize real-world concepts in a logical, machine-readable way. By carefully choosing classes, properties, and a simple axiom, I was able to model the idea of a sandwich clearly. In the future, this could be expanded to include health data, dietary tags, or even allow sandwich builders to auto-suggest ingredients.

#### **References:**

1. GitHub (2025). GitHub Copilot. [online] GitHub. Available at: <https://github.com/features/copilot> [Accessed 20 Jun 2025].
2. Simple English Wikipedia. (2023). Sandwich. [online] Available at: <https://simple.wikipedia.org/wiki/Sandwich> [Accessed 19 Jun 2025].